MILITARY SPECIFICATION

TEST SET, THERMAL SIGHT, TS-3681/VSG-2

This specification is approved for use by USAERADCOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Test Set, Thermal Sight, TS-3681/VSG-2.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	- Boxes, Folding, Paperboard
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-B-665	- Boxes, Paperboard, Metal Edged and Components
PPP-B-676	- Boxes, Setup

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MIL-P-116	- Preservation-Packaging, Methods of
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-T-28800	- Test Equipment for Use With Electrical and Electronic Equipment, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, USA Electronics Research and Development Command, ATTN: DRDEL-E, 2800 Powder Mill Road, Adelphi, MD 20783 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5855

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	- Environmental Test Methods

DRAWINGS

DLSM-B-804916	- Test Set, Sight, Thermal, Tank Mounted AN/VSG-2

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The Test Set, Thermal Sight, TS-3681/VSG-2, herein known as the test set, shall support maintenance of the Sight, Thermal, Tank Mounted, AN/VSG-2 equipment at the direct support and general support (DS/GS) maintenance environments. The test set shall consist of two containers. MIL-T-28800 shall be used to define the general requirements of the Collimator, Type I, class 4, style T, and type I, class 4, style C for the test station.

a. 58-inch collimator. Container number 1.

b. Line Replaceable Unit (LRU) test station. Container number 2.

3.2 Construction. The test set shall be constructed in accordance with DLSM-B-804916 or as specified in the contract.

3.3 First article. When specified in the contract (see 6.2), the contractor shall furnish test sets for first article inspections.

3.4 Parts, materials, and processes. Parts, materials, and processes shall be as specified herein and as shown on the applicable drawings. Parts, materials, and processes not specified shall be selected in accordance with MIL-P-11268.

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3.5 Components. The test set consists of two holding fixtures that will provide a stable mount for the TTS LRU's and two containers that house the following components. Container number one houses the collimator with its temperature controller, spindle and drum assembly and the input power cable. Container number two houses the LRU test station, test target reticle combiner, LED viewer telescope, PWB puller, spanner wrench, and the TTS interconnect cables. Each component of the two containers shall be constructed in accordance with their applicable drawings, specifications, and as specified herein.

3.6 Preconditioning. Each test set shall be subjected to a 48 hour preconditioning prior to delivery.

3.7 Operation.

3.7.1 Power.

3.7.1.1 Collimator. The collimator shall operate from a 115 Vac, 50/60 Hz single phase power source.

3.7.1.2 Test station. The LRU test station shall operate from a direct current source from a minimum of 18 volts to a maximum of 30 volts. The maximum peak current drain at any time during operation of the test station shall not exceed 25 amperes.

3.7.2 Collimator output. The collimator shall contain a temperature controller and spindle and drum assembly which when used will provide a combination of six (6) collimated infrared and visible targets to the unit under test sufficient to derive the performance status. The temperature controller and target source assembly shall create a calibrated temperature difference between its blackbody source and the target array. This temperature difference at the resolution frequency shall be 0.45, $\pm$0.045°C for the MRT targets. For the other targets, the temperature difference shall be 4.50, $\pm$0.45°C.

3.7.3 Test station electrical protection. The test station shall be equipped with devices to protect the system from reverse polarity, overcurrent, short circuit and under/over voltages. These devices shall be located on the input power side of the test station.

3.7.4 Test station output characteristics. The test station shall have two modes of operation, STANDBY and ON. In the STANDBY mode, the test station shall provide the following outputs:

a. 120, $\pm$10 Vac (under load), 400, $\pm$20 Hz, 2-phase, square wave.
b. 18 to 30 Vdc.
c. 5.0, $\pm$0.5 Vdc (for operation of BITE circuitry).

In the ON mode, all of the voltages listed for the STANDBY mode shall be provided plus the following additional voltages:

d. 18 to 30 Vdc (this in addition to the 18 to 30 Vdc in STANDBY).
e. Eight regulated voltages as shown below.

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NOTE: Input ripple voltage equals 3.0 V p-p at 0 to 35 kHz.

Output voltage (Vdc)	Maximum continuous current (amperes)	Minimum ripple rejection (dB)
+10.5 ±0.5	1.0	60
+10.0 ±0.5	1.0	50
-9.0 ±0.5	1.0	50
+10.0 +8.0 -0.5	2.0	50
+5.0 ±0.3	0.750	50
-5.0 ±0.3	0.500	50
+15.0 ±0.5	1.0	50
-15.0 ±0.5	0.750	50

3.7.5 Test target/reticle combiner. The test target/reticle combiner shall combine a visible target from the collimator with the reticle image for reticle focus. It shall also divide and separate the target, providing a double image for adjusting IR magnification.

3.7.6 LED viewer telescope. The LED viewer telescope shall allow visual evaluation of the LED image from the TTS head assembly.

3.7.7 Controls.

3.7.7.1 Collimator controls.

3.7.7.1.1 OFF/ON. The OFF/ON switch shall control application of power to the temperature controller and provide 3-ampere circuit breaker protection.

3.7.7.1.2 Target select. The target select control shall provide an operator the option to select any one of six available targets.

3.7.7.1.3 ON/ALIGN. The ON/ALIGN switch shall permit either alignment or normal operation of the temperature controller.

3.7.7.2 Test station controls.

3.7.7.2.1 Power converter simulator OFF/ON. The power converter simulator OFF/ON control shall provide the operator the capability of simulating the power converter characteristics.

3.7.7.2.2 Power converter simulator BITE. The power converter simulator BITE control shall provide a SAFE/FAIL indication of the internal power converter.

3.7.7.2.3 Reticle brightness. A reticle brightness control shall vary the intensity of the reticle brightness.

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3.7.7.2.4 Commander/gunner. The commander/gunner control shall switch command functions between commander's and gunner's displays.

3.7.7.2.5 Polarity. The polarity control shall allow operator selection of either white hot or black hot video polarity.

3.7.7.2.6 System test. The system test switch shall allow the operator to exercise the BITE circuitry of the LRU under test.

3.7.7.2.7 Brightness. A brightness control shall simulate the gunner's/commander's display controls.

3.7.7.2.8 Contrast. A contrast control shall simulate the gunner's/commander's display control.

3.7.7.2.9 Mode select. A mode-select switch shall simulate the OFF/STBY/ON switch on the gunner's display.

3.7.7.2.10 Cool/warm. A cool/warm control shall simulate the cooldown circuitry in the head assembly. This control shall be disabled when a head assembly is under test.

3.7.7.2.11 Narrow/wide field of view (FOV). A control shall simulate the selection of either NFOV or WFOV in the head assembly.

3.7.7.2.12 Dc converter load OFF/ON. This control shall allow an operator to load the dc circuits of a power converter under test to 80 percent of capacity.

3.7.7.2.13 115 Vac load OFF/ON. This control shall allow an operator to load the ac output of a power converter under test to 80 percent of capacity.

3.7.7.2.14 Reticle slew. This control shall slew the reticle in azimuth.

3.7.7.2.15 Function select. This control shall allow operator selection of a group of three signals.

3.7.7.2.16 Field select. This control shall allow selection of a single signal from the group of three selected by the function select switch.

3.7.8 Interconnect cables. The test station shall be equipped with seven interconnecting cables that shall provide the capability of performing all necessary tests and alignments on the TTS and shall provide for self testing of the LRU test station.

3.8 Environment. The test set shall operate without damage or degradation in performance during and after exposure and testing to the environmental conditions specified. Damage is defined as degradation in specified performance, breakage, loosening, shift, evidence of condensation or moisture within the system, and failure to operate in accordance with the requirements of this specification.

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3.8.1 Altitude. The test set shall operate at altitudes up to 15,000 feet above sea level and be transportable at altitudes up to 40,000 feet above sea level without damage.

3.8.2 High temperature.

3.8.2.1 Operation. The test set shall operate at temperatures to +55°C.

3.8.2.2 Storage. The test set shall withstand storage at temperatures to +85°C.

3.8.3 Low temperature.

3.8.3.1 Operation. The test set shall operate at temperatures to 0°C.

3.8.3.2 Storage. The test set shall withstand storage at temperatures to -62°C.

3.8.4 Humidity. The test set shall not be damaged by exposure to 95, (+5 -0) percent relative humidity.

3.8.5 Fungus. The test set, without its shipping container, shall not be damaged by exposure to fungus growth.

3.8.6 Salt fog.

3.8.6.1 Test station. The test station in its shipping container shall not be damaged by exposure to a salt fog atmosphere of 5 percent NaCl atomized spray for 48 hours.

3.8.6.2 Collimator. The collimator in its enclosure shall not be damaged by exposure to a salt fog atmosphere of 20 percent NaCl atomized spray for 48 hours.

3.8.7 Watertight. The collimator, with its cover on, shall not be damaged after being submerged to a depth of 3 feet in water.

3.8.8 Shock.

3.8.8.1 Shock pulse level. The test set shall operate properly after being exposed to shock inputs up to 30 g's.

3.8.8.2 Bench handling. The test set shall not be damaged by bench handling during normal maintenance activities.

3.8.8.3 Transit drop.

3.8.8.3.1 Test station. The test station shall not be damaged after being subjected to transit drops from a height of 12 inches while contained in its transit case.

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3.8.8.3.2 Collimator. The collimator shall not be damaged after being subjected to transit drops from a height of 30 inches while contained in its transit case.

3.8.9 Vibration. The test set (containers with contents installed and covers in place) shall show no degradation in appearance or performance after being subjected to vibration inputs up to 3 g's.

3.8.10 Sand and dust resistance (collimator only). The collimator, with the cover installed and latched, shall not be damaged after being exposed to dust at velocities up to 1750 feet per minute for periods up to 12 hours.

3.9 Treatment and painting. Unless otherwise specified, the test set and its components and parts shall be cleaned, treated, and painted in accordance with the applicable drawing.

3.10 Nameplates and product marking. Unless otherwise specified, the test set, parts, components, subassemblies, and assemblies thereof shall be marked for identification and reference designation markings in accordance with the applicable drawing.

3.11 Workmanship. Workmanship shall be as specified in MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.5).
c. Inspection of packaging (see 4.9).

4.3 First article inspection. Unless otherwise specified in the contract or purchase order, first article inspection shall be performed by the contractor.

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4.3.1 Inspections. Each assembled first article test set shall be examined in accordance with table I. All materials, parts, processes, and assemblies shall be examined for conformance to the applicable specification or drawing. Inspections shall be made using MIL-STD-105, General Inspection Level II with a 2.5 percent AQL for mechanical and a 4 percent AQL for visual inspection, except as noted in table I.

TABLE I. Inspections.

Inspection Requirement	Percent AQL	Requirement paragraph	Inspection paragraph
Any part or component missing or damaged	2.5	3.2, 3.4	MIL-STD-252
Marking not as specified	2.5	3.2	
Workmanship not as specified	2.5	3.11	MIL-STD-454
Treatment and painting not as specified	2.5 1/	3.2	
Controls	2.5	3.7.7	4.7.7
Dimensions not as specified	2.5	3.2	

1/ MIL-STD-105, Inspection Level S-3 or objective quality evidence.

4.3.2 Tests criteria. Upon successful completion of inspections specified in 4.3.1 and 4.5.1.1, first article test sets shall be subjected to the tests in table II. Tests shall be conducted in the order listed. Failure of any tests shall constitute first article failure.

4.3.3 Tests.

4.3.3.1 First article inspection, subgroup 1. From test sets that have passed 4.3.1 and 4.5.1.1, three (3) first article test sets shall be subjected to all tests of table II subgroup 1. Failure of any test set to pass all tests in table II, subgroup 1 shall constitute failure of first article.

4.3.3.2 Inspection procedures. Unless otherwise specified herein, the test set shall be tested at an ambient temperature of 23°C, ±5°C. All specified tolerances are absolute and errors due to test equipment shall be subtracted from these tolerances.

4.3.3.3 First article inspection, subgroup 2. Two (2) test sets which have successfully passed the tests specified in 4.3.3.1 shall be subjected to all tests of table II, subgroup 2 in any order. At the completion of subgroup 2, the tests of subgroup 1 shall be repeated. Failure of any test set to pass all tests of table II, subgroup 2 and subgroup 1 shall constitute first article failure.

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TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
Power	3.7.1	4.7.1
Collimator output	3.7.2	4.7.2
Test station electrical protection	3.7.3	4.7.3
Test station output characteristics	3.7.4	4.7.4
Test target/reticle combiner	3.7.5	4.7.5
LED viewer telescope	3.7.6	4.7.6
Controls	3.7.7	4.7.7
Interconnect cables	3.7.8	4.7.8
Subgroup 2		
Altitude	3.8.1	4.8.1
High temperature	3.8.2	4.8.2
Low temperature	3.8.3	4.8.3
Humidity	3.8.4	4.8.4
Fungus	3.8.5	4.8.5
Salt fog	3.8.6	4.8.6
Watertight	3.8.7	4.8.7
Shock	3.8.8	4.8.8
Vibration	3.8.9	4.8.9
Sand and dust	3.8.10	4.8.10

4.4 Inspection Procedures. All tolerances herein are absolute. Any test measuring equipment tolerances shall be subtracted from the test set test measurement. Unless otherwise specified, no internal adjustments or component replacement shall be made on any test set subsequent to the start of any of the tests described herein.

4.5 Quality conformance inspection.

4.5.1 Inspection. Inspection shall be as specified in 4.3.1.

4.5.1.1 Preconditioning. Each test set which has passed the inspection of 4.3.1 shall be subjected to the preconditioning of 4.6.

4.5.2 Tests.

4.5.2.1 Group A inspection. Group A inspection shall be conducted on each test set which has passed the inspection specified in 4.5.1. Group A tests listed in table III may be performed in any order. Failure of any test shall be cause for rejection of that unit.

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TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Power	3.7.1	4.7.1
Collimator output	3.7.2	4.7.2
Test station electrical protection	3.7.3	4.7.3
Test station output characteristics	3.7.4	4.7.4
Test target/reticle combiner	3.7.5	4.7.5
LED viewer telescope	3.7.6	4.7.6
Controls	3.7.7	4.7.7
Interconnect cables	3.7.8	4.7.8

4.5.2.2 Group B inspection. Group B inspection does not apply.

4.5.2.3 Group C inspection. Group C inspections shall be conducted on test sets selected from units which have passed the tests in 4.5.2.1. The sample(s) shall be tested in accordance with the inspections listed in table IV. Samples shall be selected in accordance with 4.5.2.3.1. Group C tests listed in table IV may be performed in any order.

4.5.2.3.1 Sampling for group C inspection.

4.5.2.3.1.1 Subgroup 1. For subgroup 1, one test set shall be selected from every third quality conformance inspection lot, starting with the first lot.

4.5.2.3.1.2 Subgroup 2. For subgroup 2, one test set shall be selected from every sixth quality conformance inspection lot, starting with the first lot.

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
Altitude	3.8.1	4.8.1
High temperature	3.8.2	4.8.2
Humidity	3.8.4	4.8.4
Low temperature	3.8.3	4.8.3
Watertight	3.8.7	4.8.7
Shock	3.8.8	4.8.8
Vibration	3.8.9	4.8.9
Sand and dust	3.8.10	4.8.10

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TABLE IV. Group C inspection - Continued

Inspection	Requirement paragraph	Test paragraph
Subgroup 2		
Electromagnetic radiation	3.7.8	4.7.8
Fungus	3.8.5	4.8.5
Salt fog	3.8.6	4.8.6

4.5.2.3.2 Group C failures. Actions required relative to group C failures shall be as specified in the contract or purchase order (see 6.2).

4.6 Preconditioning. Each test set shall be subjected to a minimum of 6 preconditioning cycles. Each cycle shall conform to the profile of figure 1. The temperature extremes shall be 0°C and +52°C. The test set shall be vibrated at 1 g at a nonresonant frequency between 20 Hz and 60 Hz for 10 minutes each operating hour. If a failure occurs, the test set shall be removed from this test, repaired, and returned to preconditioning. The preconditioning shall continue for at least one full cycle without additional failures after repair.

4.7 Tests.

4.7.1 Power.

4.7.1.1 Collimator. Connect the collimator power cable to connector J1 and apply 115 Vac. Set the temperature controller ON/OFF switch to ON and start stopwatch. ON indicator shall illuminate immediately and the READY indicator shall illuminate in 10 minutes or less. Failure to meet the requirements of 3.7.1.1 shall constitute failure of this test.

4.7.1.2 Test station. The test station shall be connected to an 18 to 30 Vdc, 25 A, power source. A dc voltmeter shall be used to measure the input voltage at the input connector. A dc ammeter shall be connected in series with the input line. The 120 Vac, 400 Hz, output, and the regulated dc outputs shall be loaded by placing the 115 VAC LOAD switch and the DC CONVERTER LOAD switch to ON. Set the input voltage to 30 Vdc and measure the current. The product of the input voltage and the current shall not exceed 300 watts. The test set shall be turned off and then turned back on at each of the voltage extremes to ensure the test set will start and operate at these voltages. Failure to meet the requirements of 3.7.1.2 shall constitute failure of this test.

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4.7.2 Collimator output. Verify that the temperature difference between the thermal target black body source and the target array meets the requirement of 3.7.2. Rotate the drum and spindle control through its full rotation and verify the control encounters six positive detents with a different target being presented at each detent. Position the thermal channel/visible target for display in the collimator and verify that the target light sources are illuminated. Failure to meet the requirements of 3.7.2 shall constitute failure of this test.

4.7.3 Test station electrical protection.

4.7.3.1 Reverse voltage protect. Adjust the input power to -26, ±2 Vdc and turn the power converter simulator on. The circuit breaker shall open in 5 seconds or less.

4.7.3.2 Overcurrent/short circuit protect. Apply a 10-ampere load to each regulated dc output one at a time and observe that the voltage drops below 5 volts (fold back). Remove load and observe that the voltage returns to normal. Short circuit the 115-Vac output and observe that the circuit breaker opens. Remove the short circuit, reset the circuit breaker, and observe that the 115 Vac returns to normal.

4.7.3.3 Overvoltage protect. Adjust the input power to 40.0, ±0.5 Vdc and turn the power converter simulator on. The circuit breaker shall open in 5 seconds or less. Failure to meet the requirements of 3.7.3 shall constitute failure of this test.

4.7.4 Test station output characteristics.

4.7.4.1 DC voltage test (unloaded). Using a test cable and with 26, ±2 Vdc applied, the measured dc voltages at each function select switch position shall be within the tolerance specified in 3.7.4.

4.7.4.2 DC voltage test (loaded). With the test conditions of 4.7.4.1 and the DC CONVERTER LOAD switch ON, the measured dc voltages shall not vary more than 10 percent from the results of 4.7.4.1.

4.7.4.3 AC voltage test (unloaded). With the test conditions of 4.7.4.1 the measured ac output voltage shall be 130, (+15 -7) Vac.

4.7.4.4 AC voltage test (loaded). With the test conditions of 4.7.4.1 and the AC LOAD switch ON, the ac output voltage shall be not less than 110 Vac. Failure to meet the requirement of 3.7.4 shall constitute failure of this test.

4.7.5 Test target/reticle combiner. With the test target/reticle combiner installed on a TTS gunner's display, set the IR boresight knobs to their mid-travel position and set the reticle projector feedback voltage to 0.000, ±0.010 volt. Align the left target with the reticle as viewed through the test target/reticle combiner. Manually rotate the reticle projector gear until the reticle projector feed back voltage indicates 3.40, ±0.02 volts. The reticle, as viewed through the test target/reticle combiner, shall now coincide with

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the right target within ±0.2 ballistic mil. Measure any displacement between the target and the reticle by moving the IR boresight (azimuth) knob. Failure to meet the requirements of 3.7.5 shall constitute failure of this test.

4.7.6 LED viewer telescope. The LED viewer telescope shall resolve group 0, element 1 of the 1951 US Air Force resolution pattern. Failure to meet the requirements of 3.7.6 shall constitute failure of this test.

4.7.7 Controls.

4.7.7.1 Collimator controls.

4.7.7.1.1 OFF/ON. Verify that the OFF/ON control applies power to the temperature controller when moved from the OFF to the ON position by observing the ON indicator.

4.7.7.1.2 Target select. The target select control shall rotate with six positive detents. Check that stops are provided at both ends of travel.

4.7.7.1.3 ON/ALIGN. Connect a DMM to A1J9(+) and A1J7(GND) place the ON/ALIGN switch to align and measure 4.30, ±0.05 Vdc. Move the ON/ALIGN switch to ON and observe the presence of a voltage. Failure to meet the requirements of 3.7.7 shall constitute failure of this test.

4.7.7.2 Test station controls. Each control shall be tested for operation and conformance as specified in 3.7.7.2. Failure of each control to function as specified shall constitute failure of this test.

4.7.8 Interconnect cables.

4.7.8.1 Interconnect cables. W1, W2, W3, W4, and W7 shall be measured for continuity, conductor to conductor shorts and connector to conductor short.

4.7.8.2 Video cable W5 shall be connected to the test set connector J7. The probe end shall be connected to the vertical input of an oscilloscope. The output shall be 2, ±0.5 Vac p-p. Depress the PRESS FOR POST AMP switch, and the output shall be 15, ±2 Vdc above ground.

4.7.8.3 High voltage power supply cable W6. With the test set FUNCTION SELECT switch set to HV TEST, the FIELD SELECT switch set to position A, DMM probes connected to ALL OTHER TESTS jacks, high voltage cable W6 connected to J6, ground clip to chassis, connect the high voltage probe to an I^2 tube power supply high voltage connector. The measured voltage shall be +5.0, ±0.5 Vdc.

4.8 Environmental tests.

4.8.1 Altitude (non-operating and operating). The test shall be conducted in accordance with MIL-STD-810, Method 500, Procedure I, and except as modified herein or by MIL-T-28800. Failure to meet the requirements of 3.8.1 shall constitute failure of this test.

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4.8.2 High temperature. The test shall be conducted in accordance with MIL-STD-810, Method 501 and Method 502, Procedure I, except as modified herein or by MIL-T-28800. Failure to meet the requirements of 3.8.2 shall constitute failure of this test.

4.8.3 Low temperature. The test shall be conducted in accordance with MIL-STD-810, Method 501 and Method 502, Procedure I except as modified herein or by MIL-T-28800. Failure to meet the requirements of 3.8.3 shall constitute failure of this test.

4.8.4 Humidity. The test shall be conducted in accordance with MIL-STD-810, Method 507, Procedure IV except as modified herein or by MIL-T-28800. Failure to meet the requirements of 3.8.4 shall constitute failure of this test.

4.8.5 Fungus. The test shall be conducted in accordance with MIL-STD-810, Method 508, Procedure I, except as modified herein or by MIL-T-28800. Failure to meet the requirements of 3.8.5 shall constitute failure of this test.

4.8.6 Salt exposure. The test shall be conducted in accordance with MIL-STD-810, Method 509, Procedure I, except as modified herein or by MIL-T-28800. Failure to meet the requirements of 3.8.6 shall constitute failure of this test.

4.8.7 Watertight (collimator). The test shall be conducted in accordance with MIL-STD-810, except as modified herein or by MIL-T-28800. Failure to meet the requirements of 3.8.7 shall constitute failure of this test.

4.8.8 Shock.

4.8.8.1 Shock pulse level. The test shall be conducted in accordance with MIL-STD-810, Method 516, Procedure I except as modified herein or by MIL-T-28800. Failure to meet the requirements of 3.8.8.1 shall constitute failure of this test.

4.8.8.2 Bench handling. The test shall be conducted in accordance with MIL-STD-810, Method 516, Procedure V, except as modified herein or by MIL-T-28800. Failure to meet the requirements of 3.8.8.2 shall constitute failure of this test.

4.8.8.3 Transit drop (test station and collimator). The test shall be conducted in accordance with MIL-STD-810, Method 516, Procedure II, except as modified herein or by MIL-T-28800. Failure to meet the requirements of 3.8.8.3 shall constitute failure of this test.

4.8.9 Vibration limits. The test shall be conducted in accordance with MIL-STD-810, Method 514, Procedure X, except as modified herein or by MIL-T-28800. Failure to meet the requirements of 3.8.9 shall constitute failure of this test.

4.8.10 Sand and dust resistance (collimator). The test shall be conducted in accordance with MIL-STD-810, Method 510, Procedure I, except as modified herein or by MIL-T-28800. Failure to meet the requirements of 3.8.10 shall constitute failure of this test.

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4.9 Inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging.

5.1.1 Level A, defined in MIL-E-17555.

5.1.1.1 Cleaning. Each item of the test set listed in 3.1 shall be thoroughly cleaned as described in Method C-1 of MIL-P-116.

5.1.1.2 Drying. Each item of the test set listed in 3.1 shall be dried by an applicable procedure listed in MIL-P-116.

5.1.1.3 Unit packaging. The test set shall be individually packaged in accordance with MIL-P-116, Method III, without the use of contact preservatives.

5.1.1.4 Intermediate packaging. Unit packages shall be overpacked in close-fitting intermediate containers conforming to PPP-B-566, Class 2 of PPP-B-636, PPP-B-665, or PPP-B-676. The gross weight of intermediate containers shall not exceed 100 pounds.

5.1.2 Level C, as defined in MIL-E-17555. Each test set shall be cleaned, dried, preserved, and packaged to afford protection from the supply source to the first receiving activity in accordance with the supplier's commercial practice.

5.2 Packing. The items of the test set shall be placed in the containers. The container clasps shall be closed and two wire and lead tamperproof seals shall be affixed to each container. One of the seals shall be affixed to each container through a clasp so that the clasp cannot be released without destroying the seal. The containers shall be overpacked for shipment.

5.2.1 Level C, as defined in MIL-E-17555. The test set shall be packed in a manner that will ensure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Uniform Freight Classification Rules, or regulations of other carriers, as applicable to the mode of transportation.

5.3 Marking. In addition to any special markings required by the contract or order, unit packages, intermediate packages, and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The test set is intended for use with the Sight, Thermal Tank Mounted, AN/VSG-2.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When first article tests are to be conducted and the number of test sets in each subgroup (see 3.3, 4.3, and table II).
c. Time frame required for submission of first article test reports.
d. Time frame required for approval of first article test reports by the contracting officer.
e. Production delivery schedule defined in terms of lots for group C sampling (see 4.5.2.3).
f. Level A, level B or commercial preservation and packing (see Section 5).
g. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.
h. Actions required relative to group C failures (see 4.5.2.3.2).
i. MIL-STD-810B shall be used for environmental tests of 4.8.

CUSTODIAN:
Army - ER

PREPARING ACTIVITY:
Army - ER

Project 5855-A165

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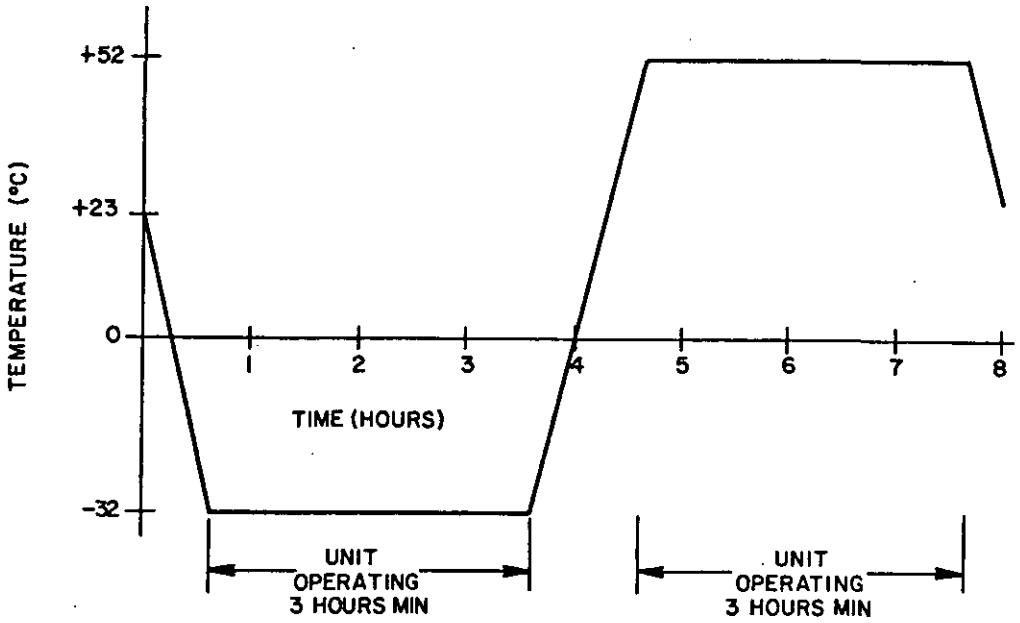


FIGURE 1. ONE PRE-CONDITIONING CYCLE.

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FOLD

POSTAGE AND FEES PAID
DOD 314

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

COMMANDER
USAERADCOM, NV&EOL
ATTN: DELNV-C

FOLD

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-T-49381(ER)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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